WISHCRAFT MASTERING LIMITER

User Manual

Version 1.0.0

A selective-clipping mastering limiter with a genuine true-peak safety stage — built to take the place of a separate clipper plugin in front of your limiter.

Concept, design, and specification by Glenn Burgos

What This Is

Wishcraft Mastering Limiter eliminates the need for a separate clipper in front of your limiter. Its Selective Clipper can be pushed harder than a standard clipper without the usual audible cost, taking real load off the limiter before it ever sees the signal. A True Peak Limiter stage then guarantees the final output stays under a genuine true-peak target, using smooth gain reduction rather than a hard clip.

Signal Path

Selective Clipper → Input Gain (Drive) → Limiter → True Peak Limiter → Safety Clip → Output

Input Gain sits after the Selective Clipper and before the Limiter — it's the “drive into the limiter” control, not a plain input trim.

Recommended Workflow

1. **Selective Clipper** — set Threshold and Selectivity first.
2. **Gain** — use Input Gain to drive the already-clipped signal into the Limiter.
3. **Limiter** — dial in Threshold, Release, Link, and Auto Gain last.

This is the opposite order from most limiter plugins, where you'd normally start with the limiter itself — here, the Selective Clipper does a lot of the work before the signal ever reaches the Limiter.

Why It's Different

The Selective Clipper only touches genuinely brief peaks — anything under about 5 ms. Sustained loud passages are left alone entirely and handled downstream by the Limiter's own gain reduction instead. That's why it can run more aggressively than a standard clipper without sounding like distortion: it's shaving isolated spikes, not flattening whole passages.

Pushed to a similar loudness, the Selective Clipper typically preserves noticeably more dynamic range (LRA) than a standard clipper. The denser and more sustained the source material, the more effective it is — on already-loud, densely-produced material the gap can be substantial; on more dynamic, transient-heavy material it narrows, since there's less sustained content for the duration gate to exempt in the first place.

Adjusting Values

Action	Effect
Click-drag	Knobs respond to vertical movement (up increases, down decreases); sliders track the mouse directly.
Shift + drag	Finer resolution — the same drag covers a smaller range of the control, for precise adjustments.
Ctrl/Cmd + click	Opens a text box to type an exact value directly.
Double-click	Resets the control to its default value.

Installation (macOS)

Copy the plugin bundles to the standard plugin folders and rescan plugins in your DAW:

Format	Location
VST3	~/Library/Audio/Plug-Ins/VST3/ (or /Library/Audio/Plug-Ins/VST3/ for all users)
Audio Unit (AU)	~/Library/Audio/Plug-Ins/Components/ (or /Library/Audio/Plug-Ins/Components/ for all users)

A Standalone application is also included and needs no installation — just run it directly.

Control Reference

Selective Clipper

Control	Range	Default	What it does
Threshold	–24 to 0 dB	–3.0 dB	Level above which the clipper can act. Lower values let it engage on quieter peaks.
Selectivity	0–100%	50%	How aggressively the duration gate exempts brief peaks. 0% = Transparent, 100% = Aggressive.
Delta	On/Off	Off	Solos what the Selective Clipper is removing, gained up so it's audible. For dialing in Threshold and Selectivity by ear — turn off before rendering.
Delta Trim	–12 to +12 dB	0.0 dB	Only shown while Delta is on. Delta's boost can be startlingly loud with aggressive clipping — trim it down here. 0.0 dB leaves Delta's boost unchanged.

Gain

Control	Range	Default	What it does
Input Gain	0–24 dB	0.0 dB	Drives the already-clipped signal into the Limiter. Sits after the Selective Clipper, before the Limiter — not a plain input trim.

Shaping EQ

Control	Range	Default	What it does
Low Shelf	–6 to +6 dB	0.0 dB	Shapes what triggers the Selective Clipper and the Limiter below ~300 Hz.
High Shelf	–6 to +6 dB	0.0 dB	Same, above ~300 Hz.

Unlike a compressor's sidechain EQ, the Limiter's copy of these filters audibly colors the output too — treat both shelves as tone controls, not something silent happening behind the scenes.

Limiter

Control	Range	Default	What it does
Threshold	–18 to 0 dB	–1.0 dB	The Limiter's gain-reduction target. A smoothed, program-dependent value the signal can transiently overshoot — not a hard-enforced ceiling. True-peak safety is the True Peak Limiter's job.
Lookahead	0.5–20 ms	3.0 ms	How far ahead the Limiter looks to smooth its gain reduction in before a peak arrives. Changing it triggers a brief engine reset, same as changing Oversampling.
Release	0–100%	30%	Program-dependent release speed, blended between a fast and a slow time constant depending on recent gain-reduction history.
Link	0–100%	75%	Stereo linking of the two channels' gain reduction, blended after smoothing using min-gain-wins logic.
TP Limit	–3 to 0 dB	–1.0 dB	The True Peak Limiter's target ceiling — see “True Peak Accuracy” below for what this guarantee actually covers.
Auto Gain	On/Off	Off	Keeps output level roughly matched regardless of gain reduction, for A/B'ing settings without a loudness bias. Strictly cut-only — never boosts. Turn off before your final render/bounce.

Utility

Control	Range	Default	What it does
Oversampling	2x / 4x / 8x	4x	Higher factors improve true-peak detection accuracy at the cost of CPU. See “Choosing an Oversampling Factor” below.
Bypass	On/Off	Off	Full latency-compensated bypass — compares the true, unprocessed input against your processed output.

Metering

Meter	Shows
Selective Clip	How much the clipper is removing, per channel (held peak value, 1-second hold with 20 dB/sec fall).
Gain Reduction	How hard the Limiter is working, per channel.
Output	Final true peak level, per channel, plus Dynamic Range (LRA) and short-term LUFS. The dot beside each channel turns yellow past your TP Limit, red past true 0 dBFS.

Technical Notes

Choosing an Oversampling Factor

2x is the lightest setting; 4x (the default) is a good balance of accuracy and CPU use for most sessions. 8x gives the True Peak Limiter the finest-resolution peak estimate and is recommended for final masters on demanding material, particularly dense, loud, transient-heavy content — CPU headroom is ample on any reasonably current machine.

Latency

The plugin reports its total processing latency to your host, which compensates for it automatically in any PDC-aware DAW. Latency depends on the Selective Clipper's internal budget, the Limiter's Lookahead setting, and the True Peak Limiter's own fixed lookahead — it does **not** depend on the Oversampling factor. Changing Lookahead updates the reported latency immediately.

True Peak Accuracy

The True Peak Limiter uses smooth gain reduction, not a hard clip, specifically because hard-clipping in the oversampled domain has been measured to still overshoot on reconstruction — the same failure mode any hard-clip-based “true peak” limiter has. On realistic program material, TP Limit is held to within roughly 0.5 dB. As with any true-peak limiter, an extreme synthetic test signal (a sustained full-scale tone parked right at the edge of Nyquist) can defeat any oversampled true-peak estimator by a wider margin — this is a known, industry-wide limitation of the technique (ITU-R BS.1770's own true-peak specification documents a comparable-order-of-magnitude bound), not something unique to this plugin. For critical delivery on unusually dense or loud transient masters, verify the final render with a dedicated true-peak meter, the same practice recommended for any true-peak limiter.

Credits

Wishcraft Mastering Limiter — concept, design, and specification by **Glenn Burgos**.

Built with the **JUCE** framework.